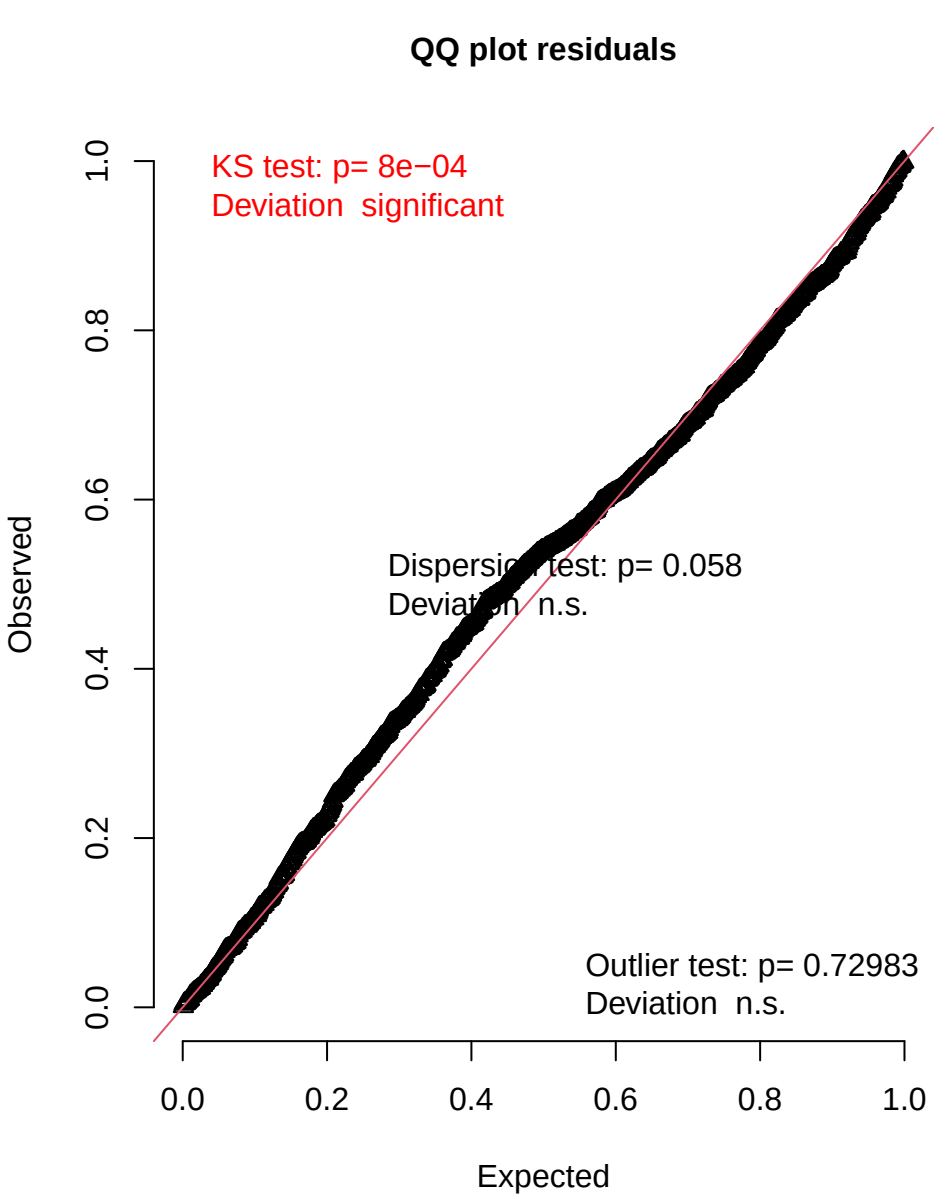
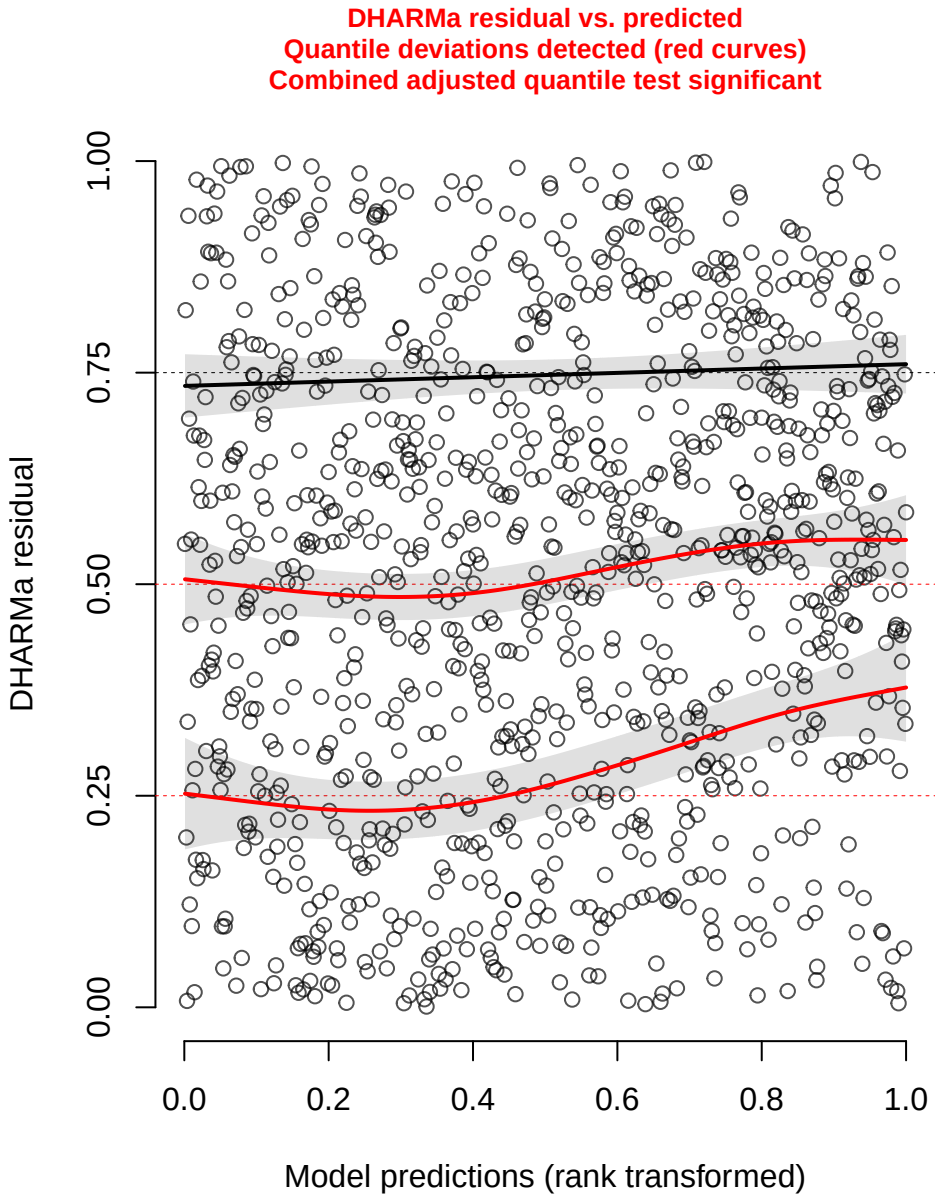
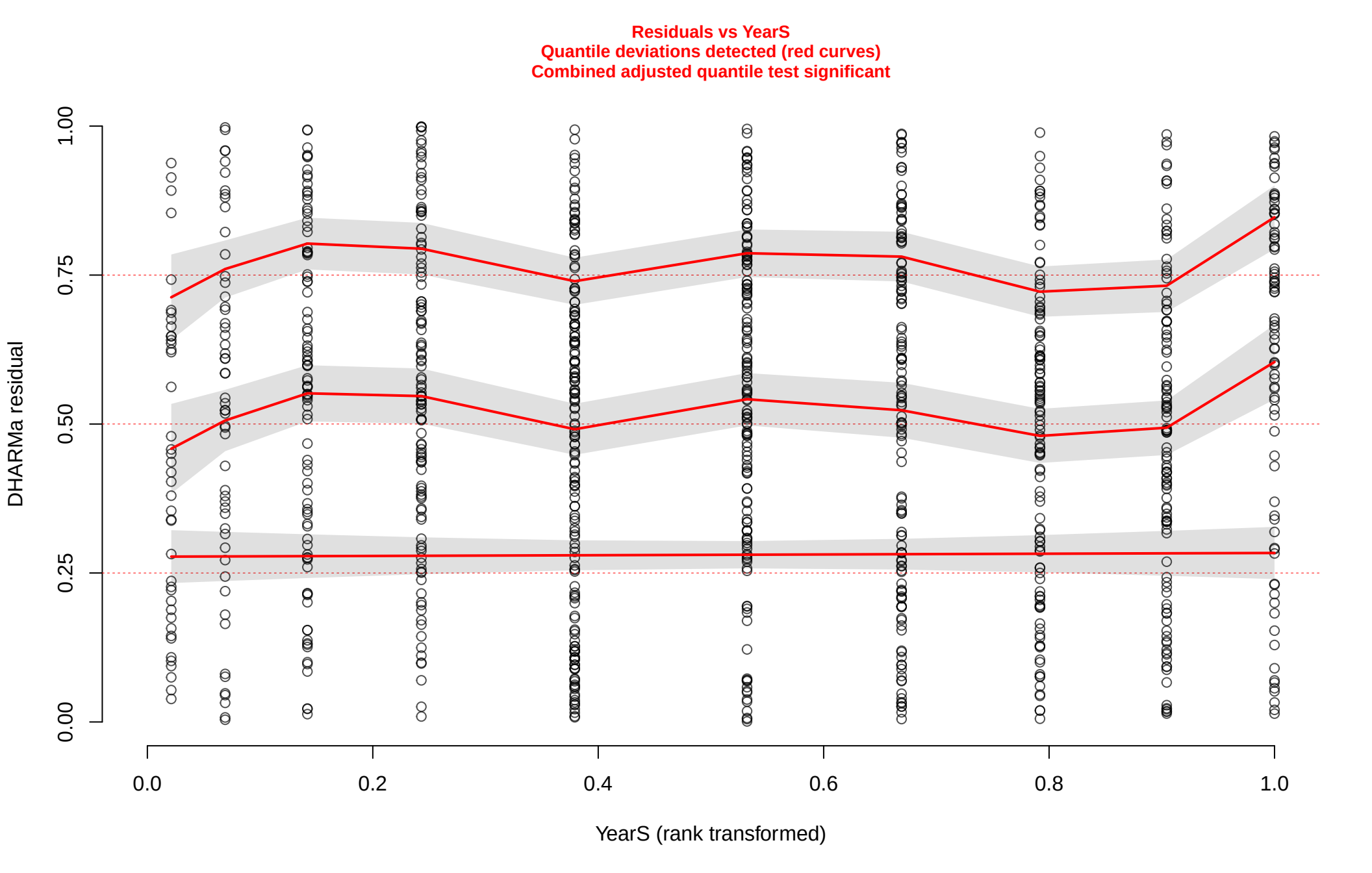
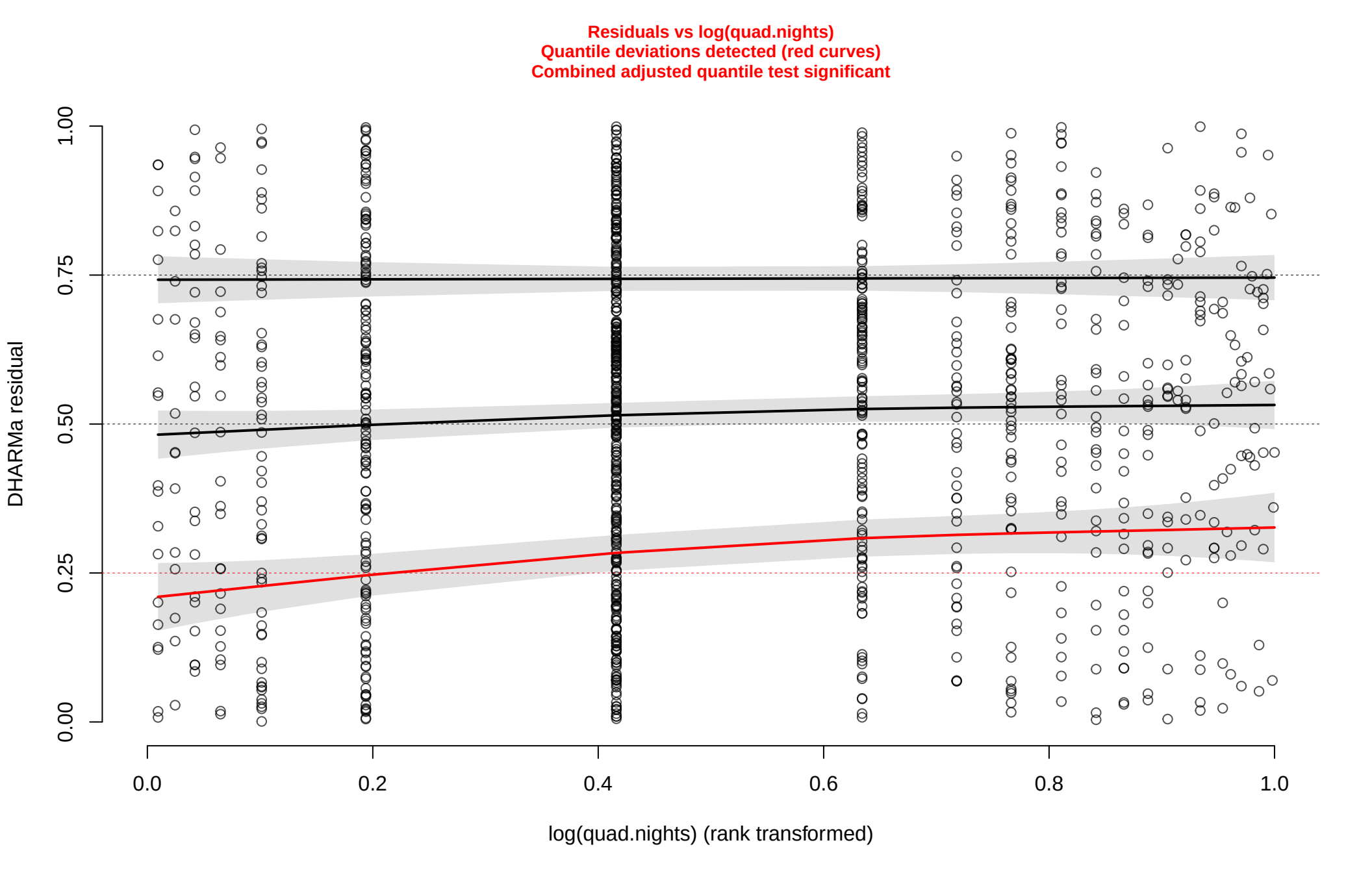
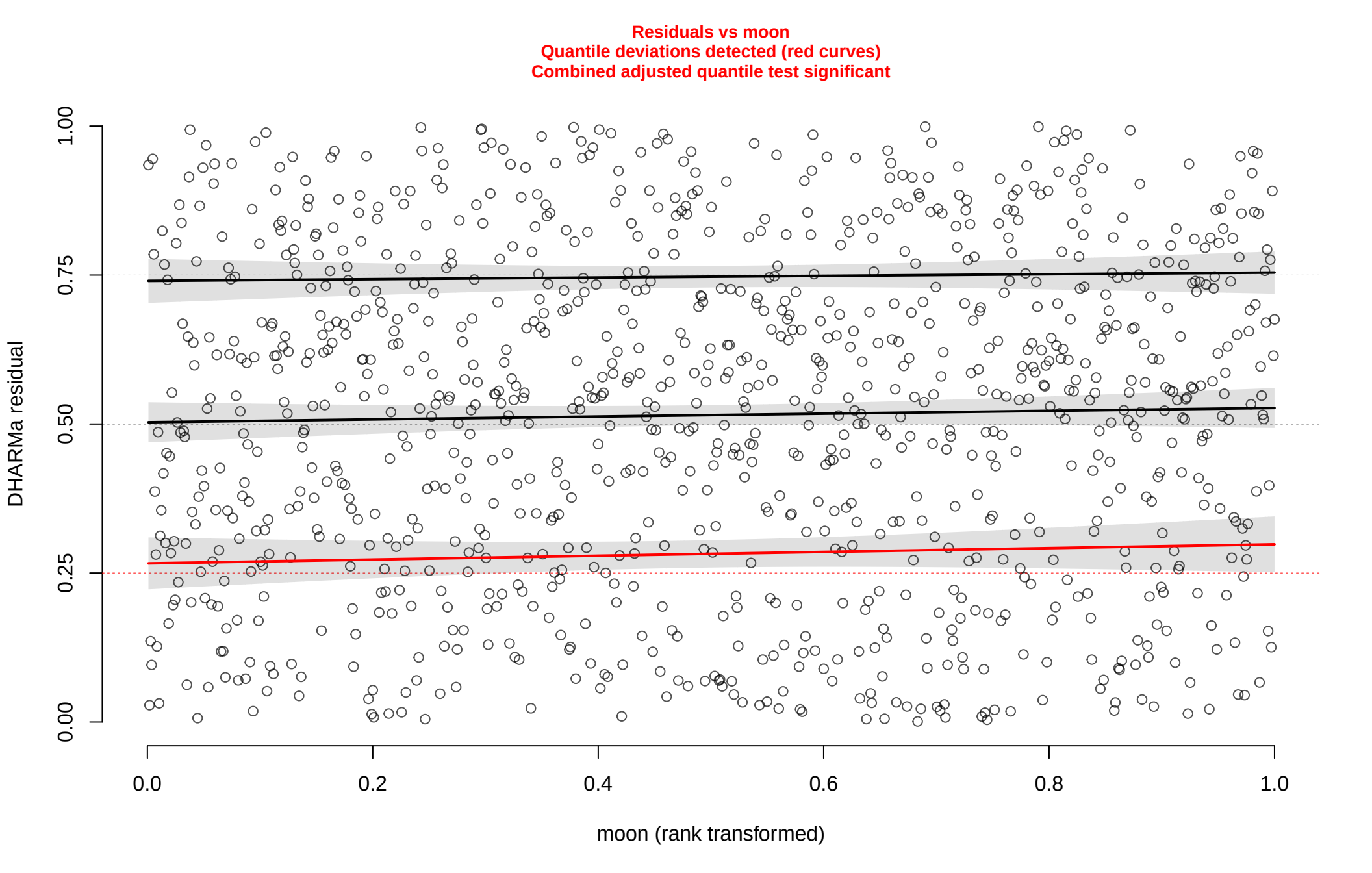
DHARMa residuals: LACI

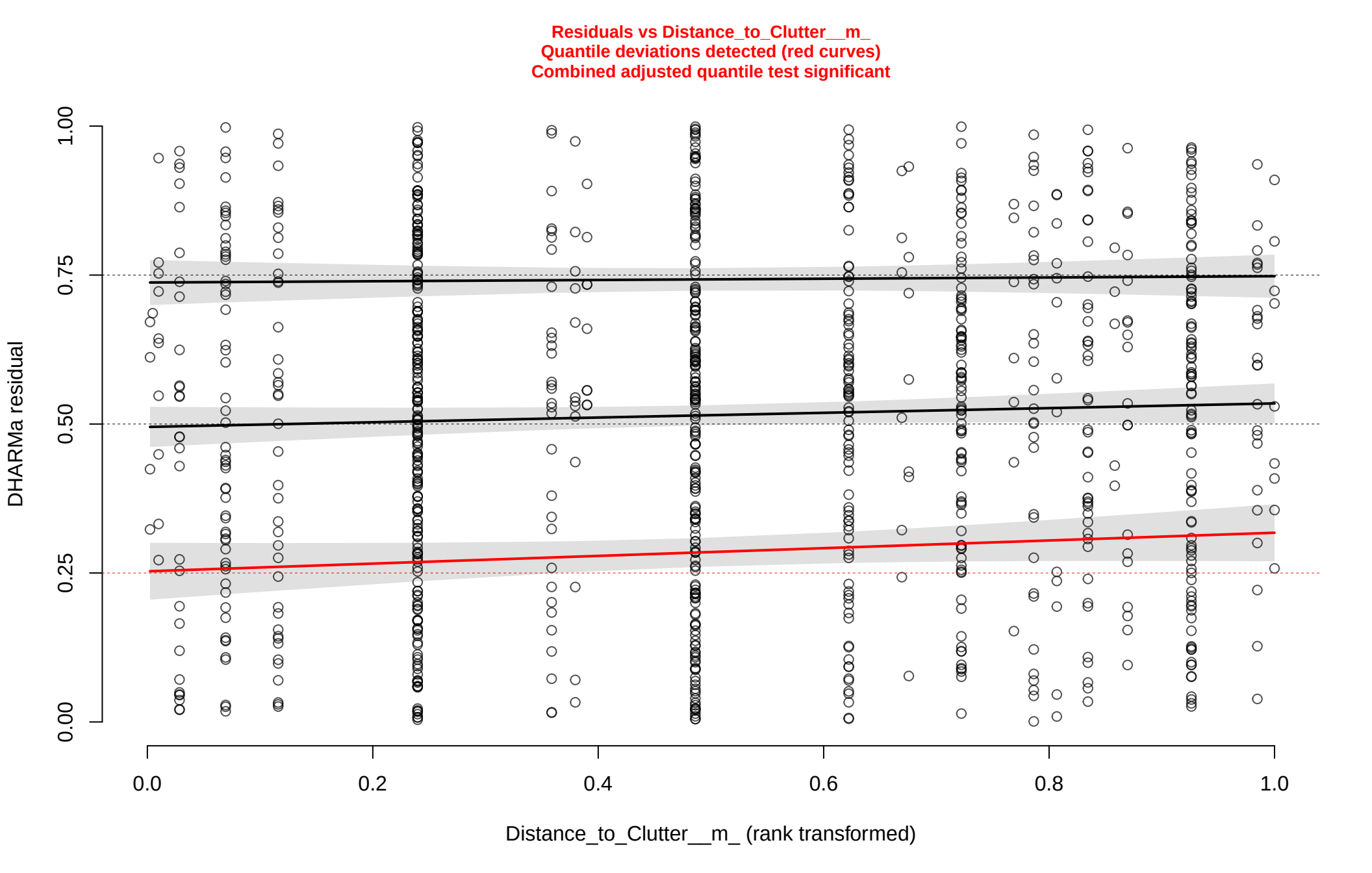


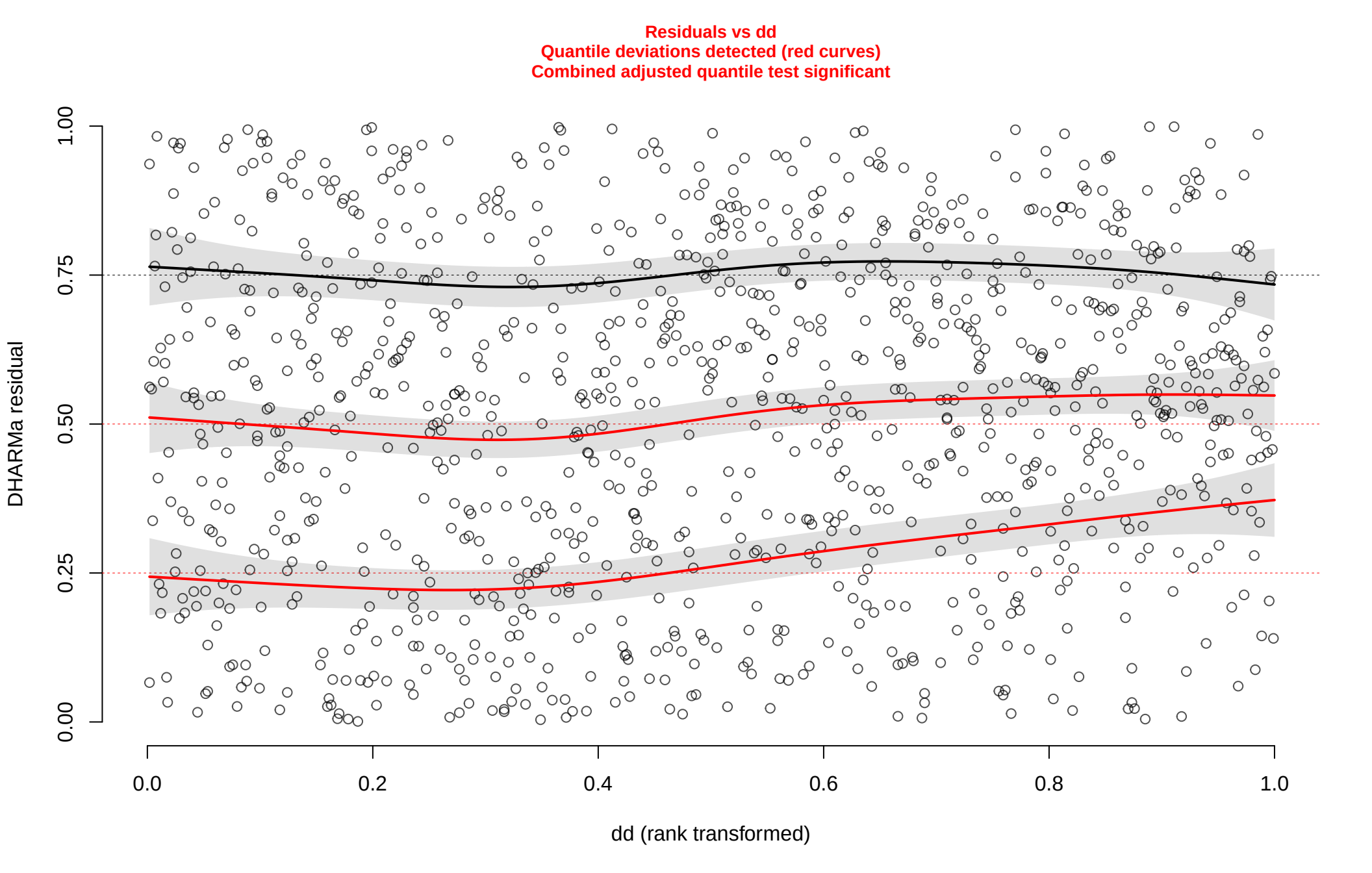


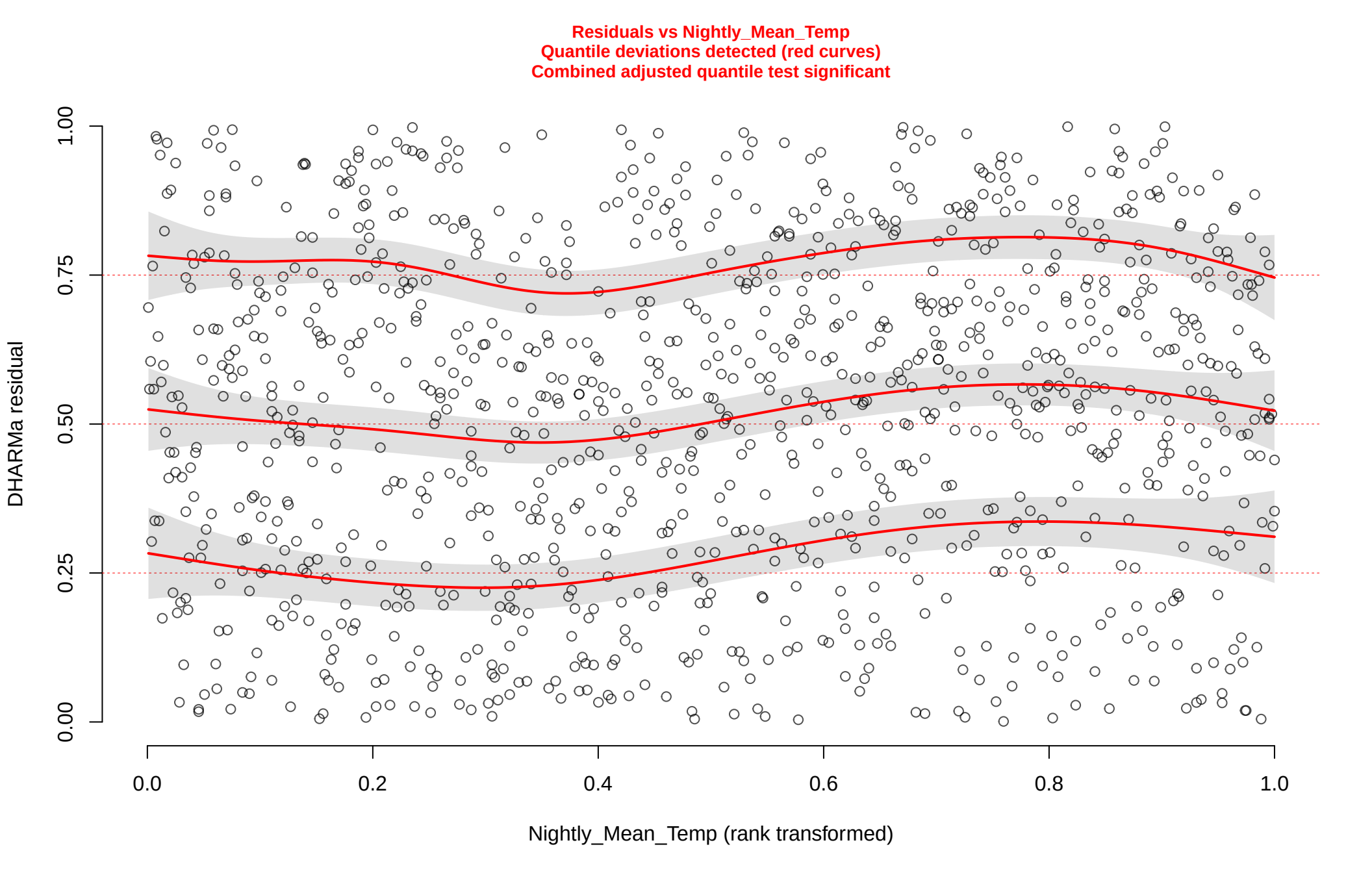

Residuals vs YearS
Quantile deviations detected (red curves)
Combined adjusted quantile test significant
DHARMa residual
1.00
0.75
0.50
0.25
0.00
0.0
0.2
0.4
0.6
0.8
1.0
YearS (rank transformed)

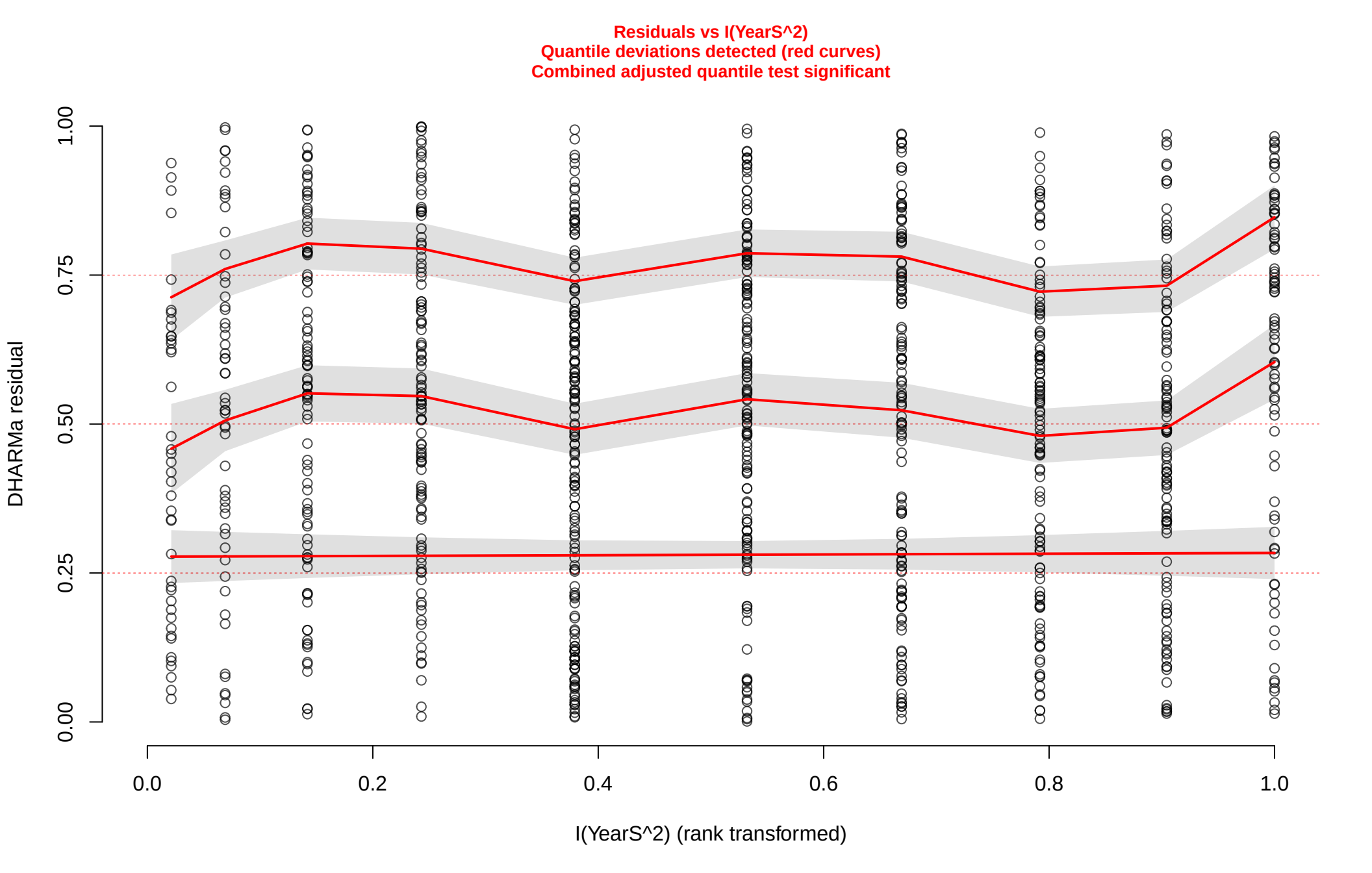

Residuals vs log(quad.nights)
Quantile deviations detected (red curves)
Combined adjusted quantile test significant
DHARMa residual
1.00
0.75
0.50
0.25
0.00
0.0
0.2
0.4
0.6
0.8
1.0
log(quad.nights) (rank transformed)

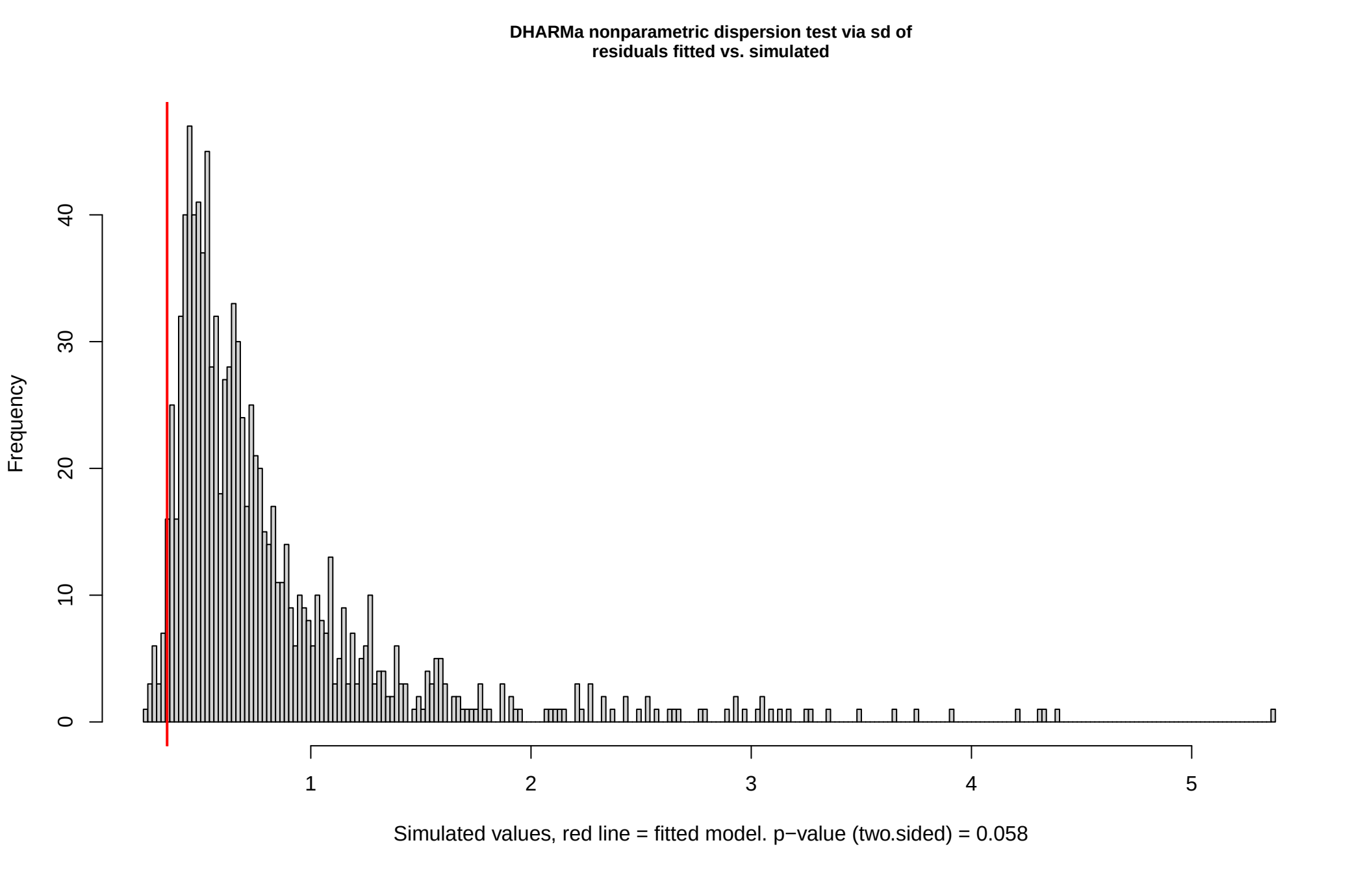

Residuals vs moon
Quantile deviations detected (red curves)
Combined adjusted quantile test significant
DHARMa residual
1.00
0.75
0.50
0.25
0.00
0.0
0.2
0.4
0.6
0.8
1.0
moon (rank transformed)

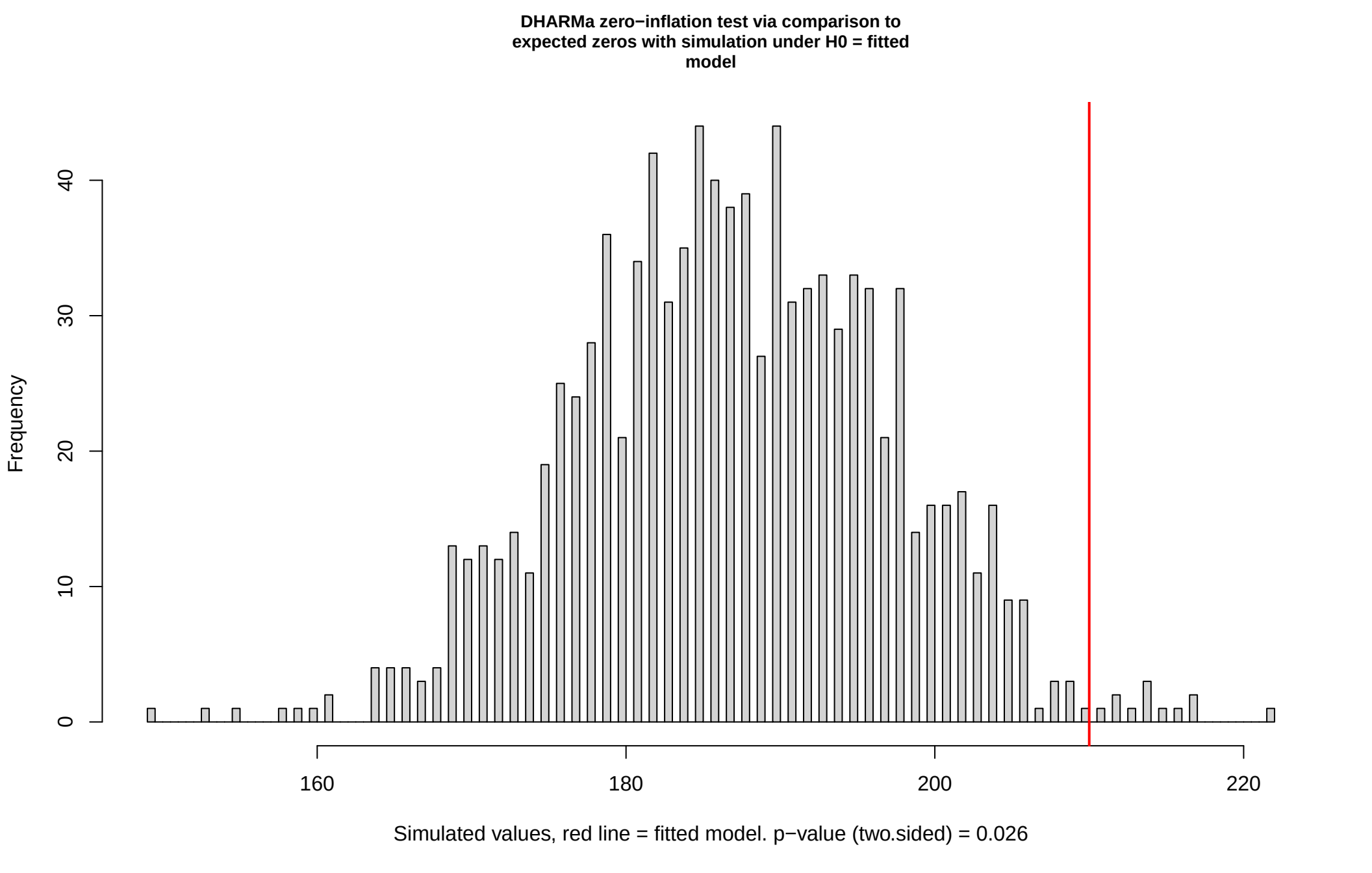

Residuals vs Distance_to_Clutter__m_
Quantile deviations detected (red curves)
Combined adjusted quantile test significant
DHARMa residual
1.00
0.75
0.50
0.25
0.00
0.0
0.2
0.4
0.6
0.8
1.0
Distance_to_Clutter__m_ (rank transformed)

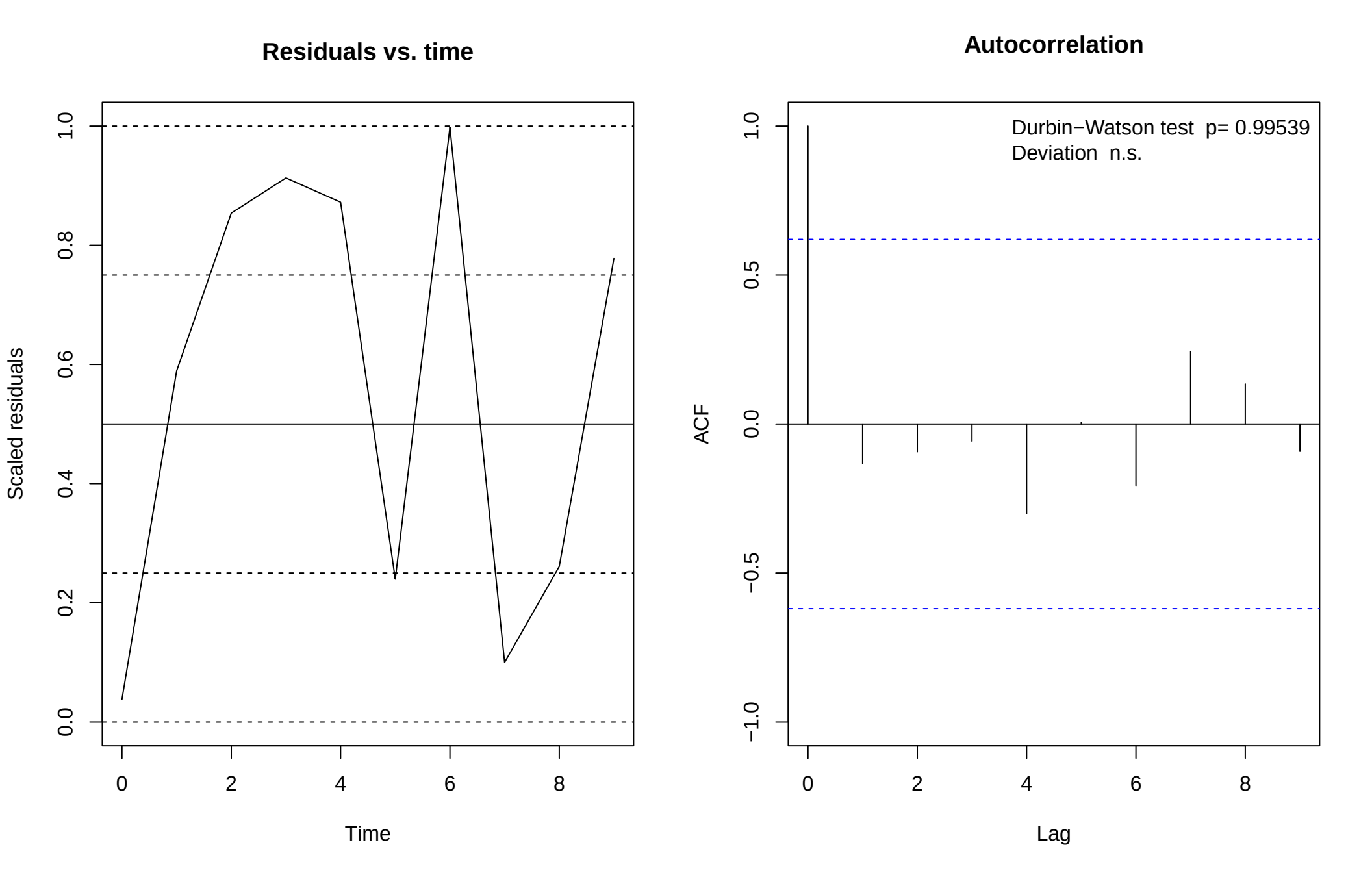

Residuals vs dd
Quantile deviations detected (red curves)
Combined adjusted quantile test significant
DHARMa residual
1.00
0.75
0.50
0.25
0.00
0.0
0.2
0.4
0.6
0.8
1.0
dd (rank transformed)

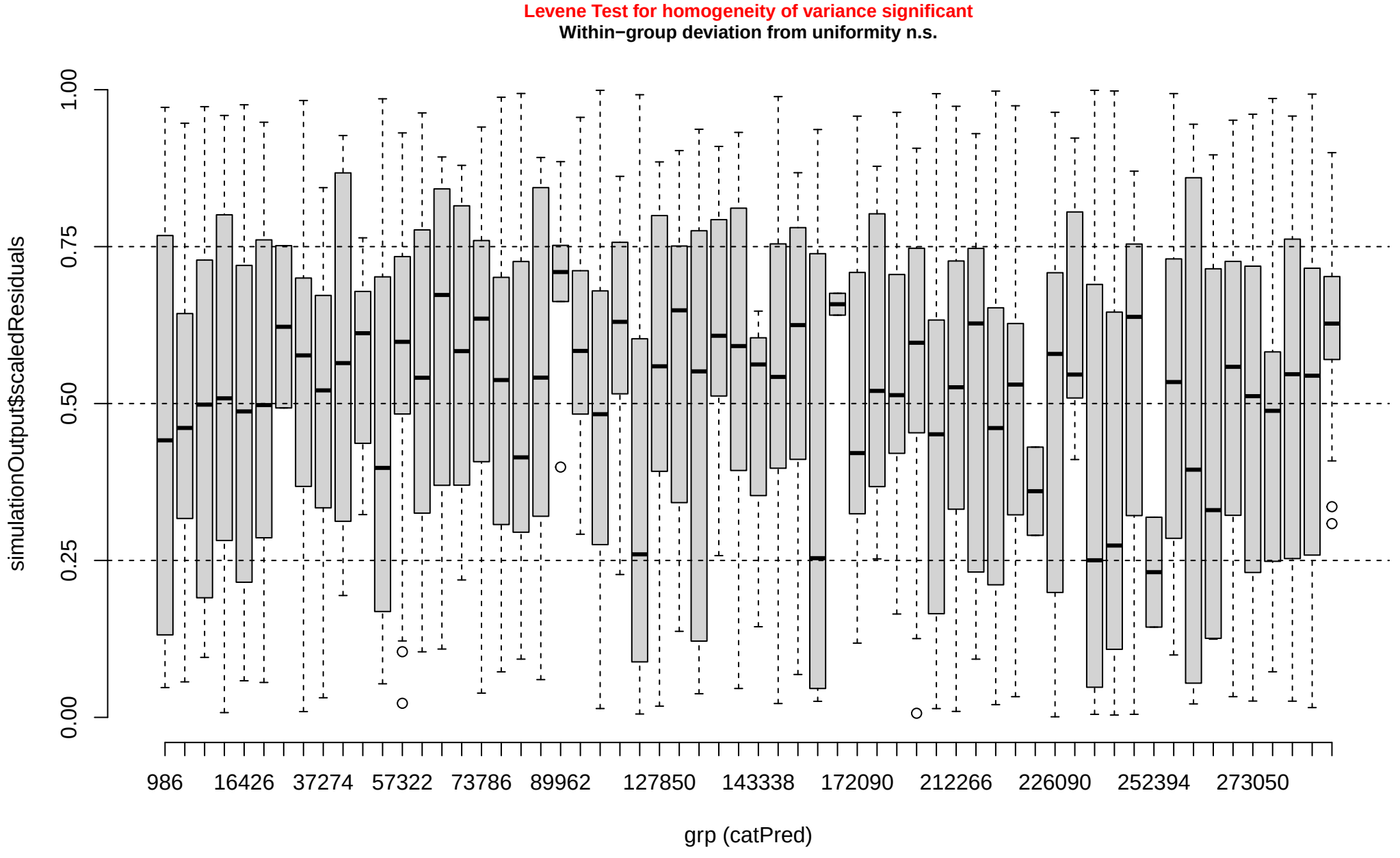

Residuals vs Nightly_Mean_Temp
Quantile deviations detected (red curves)
Combined adjusted quantile test significant
DHARMa residual
1.00
0.75
0.50
0.25
0.00
0.0
0.2
0.4
0.6
0.8
1.0
Nightly_Mean_Temp (rank transformed)

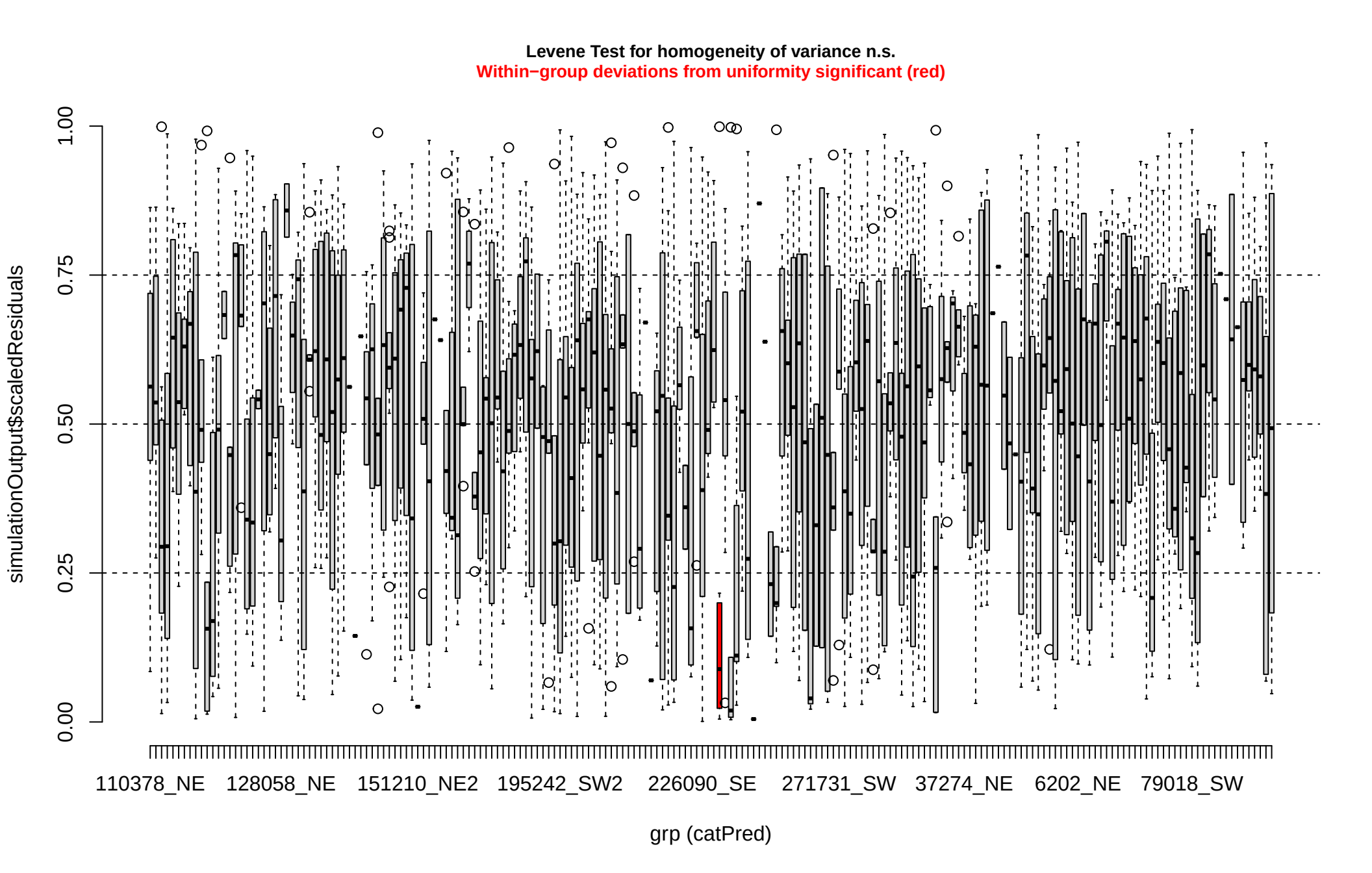

Residuals vs I(YearS^2)
Quantile deviations detected (red curves)
Combined adjusted quantile test significant
DHARMa residual
1.00
0.75
0.50
0.25
0.00
0.0
0.2
0.4
0.6
0.8
1.0
I(YearS^2) (rank transformed)

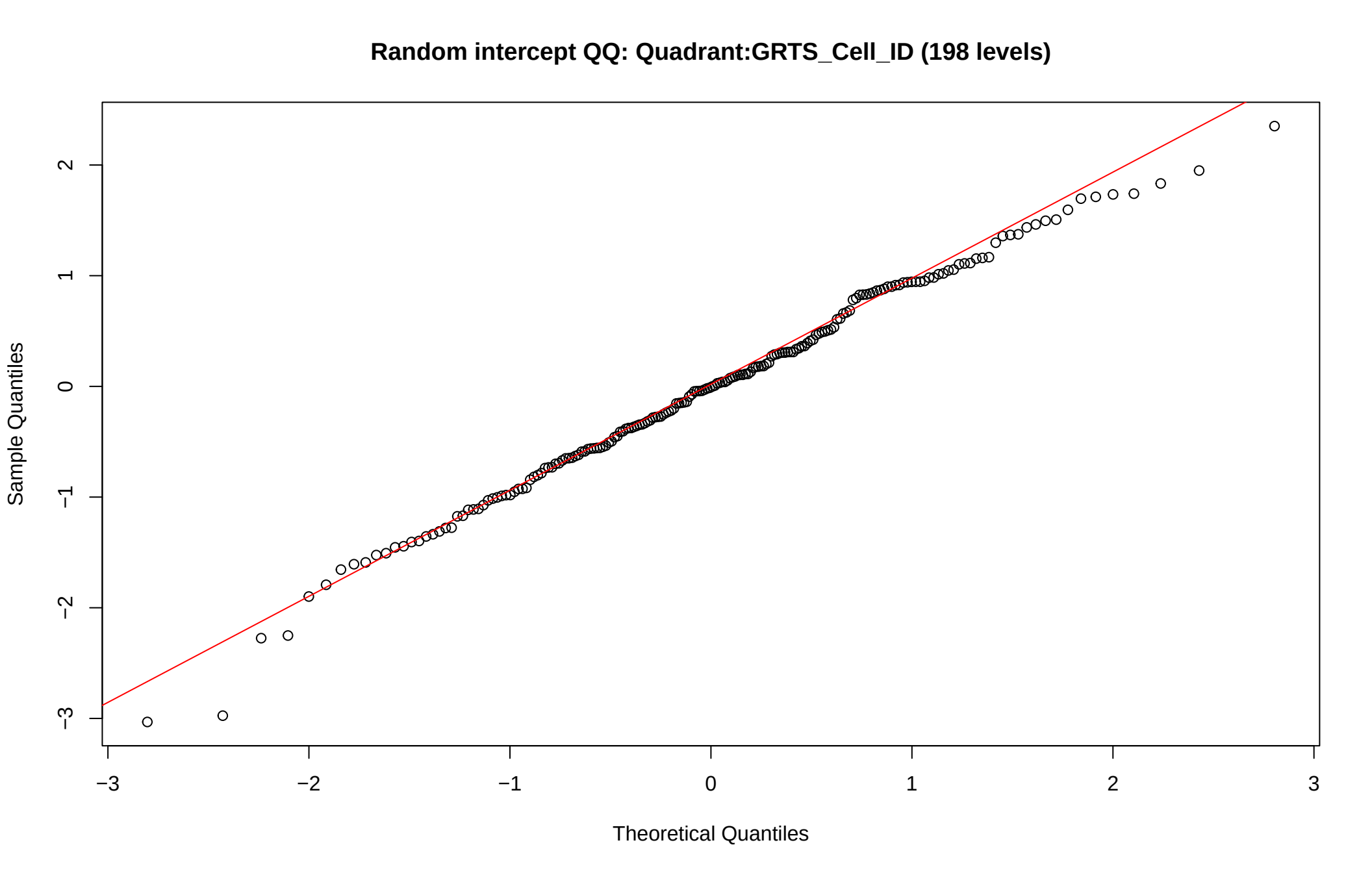

DHARMa nonparametric dispersion test via sd of residuals fitted vs. simulated
Frequency
0
10
20
30
40
1
2
3
4
5
Simulated values, red line = fitted model. p−value (two.sided) = 0.058

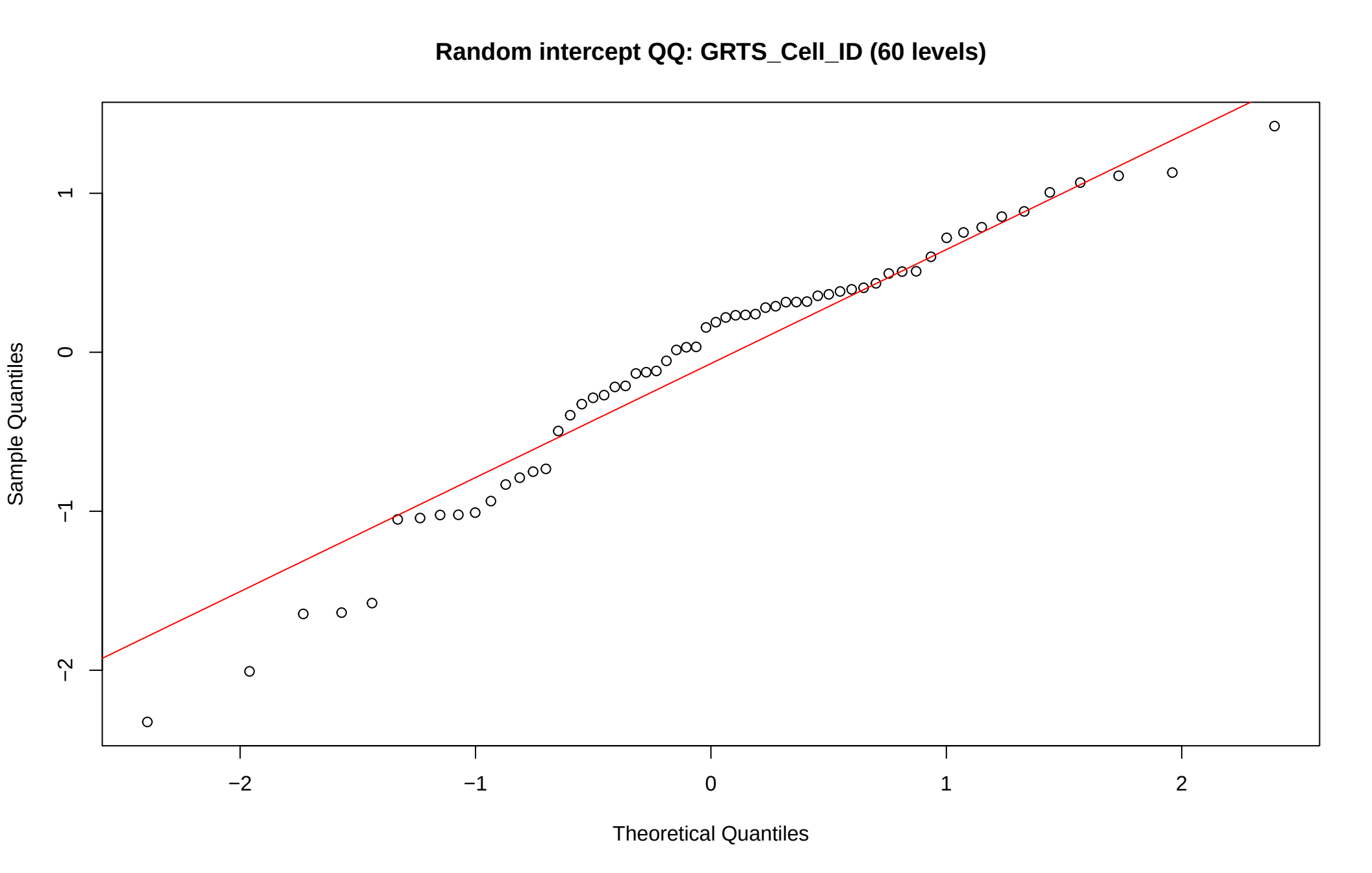

DHARMa zero−inflation test via comparison to expected zeros with simulation under H0 = fitted model
Frequency
0
10
20
30
40
160
180
200
220
Simulated values, red line = fitted model. p−value (two.sided) = 0.026


Residuals vs. time
Scaled residuals
0.0
0.2
0.4
0.6
0.8
1.0
0
2
4
6
8
Time
Autocorrelation
ACF
−1.0
−0.5
0.0
0.5
1.0
Durbin−Watson test p= 0.99539
Deviation n.s.
0
2
4
6
8
Lag


Levene Test for homogeneity of variance significant
Within−group deviation from uniformity n.s.
simulationOutput$scaledResiduals
0.00
0.25
0.50
0.75
1.00
986
16426
37274
57322
73786
89962
127850
143338
172090
212266
226090
252394
273050
grp (catPred)


Levene Test for homogeneity of variance n.s.
Within−group deviations from uniformity significant (red)
simulationOutput$scaledResiduals
1.00
0.75
0.50
0.25
0.00
110378_NE
128058_NE
151210_NE2
195242_SW2
226090_SE
271731_SW
37274_NE
6202_NE
79018_SW
grp (catPred)


Random intercept QQ: Quadrant:GRTS_Cell_ID (198 levels)
Sample Quantiles
Theoretical Quantiles
−3
−2
−1
0
1
2
3


Random intercept QQ: GRTS_Cell_ID (60 levels)
Sample Quantiles
Theoretical Quantiles
−2
−1
0
1
2
−2
−1
0
1